

JKBOSE 2026

CLASS 11TH

Bhauhanra
ON TOP.

CLASSIFICATION OF ELEMENTS AND PERIODICITY IN PROPERTIES

LECTURE-5 CHEMISTRY

VISHAL SIR



JK Bole Wallah

Topics To Covered

1 MODERN PERIODIC TABLE PART 3

REVISION.

(Metals)

S-block. (ns^{1-2})

①

H

②

Li

Be

Na

Mg

K

Ca

Rb

Sr

Cs

Ba

Fr

Ra

d-block.

3

12.

13

14

15

16

17

He.

B

C

N

O

F

Ne.

Al

Si

P

S

Cl

Ar

Ga

Ge

As

Se

Br

Kr

In

Sn

Sb

Te

I

Xe

Tl

Pb

Bi

Po

At

Rn

Nh

Fl

Mc

Lv

Ts

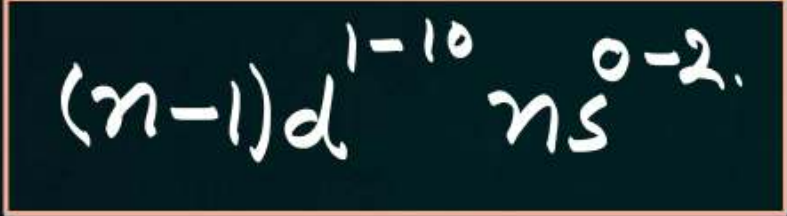
Og

$ns^2 np^{1-6}$

P-block. 18

3) d-block Elements $\Rightarrow$ Elements in which last electron enters in d-orbital of $(n-1)$ shell (penultimate).

• General Electronic Configuration.



• Includes group no. 3 to group no. 12.
 & Period no. 4, 5, 6 & 7.

These elements are also called as Transition elements. (Zn, Cd, Hg). because, there occurs a change of property (metallic to non-metallic), through them.

• Includes four series.

1st Transition series (3d series).

from Sc ($Z=21$) to Zn (30).

2nd Transition series (4d series).

from Y ($Z=39$) to Cd (48).

3rd Transition series (5d series).

from La ($Z=57$) to Hg (80).

4th Transition series (6d-series).

from Ac ($Z=89$) to Cn (112).

	3	4	5	6	7	8	9	10	11	12
1st transition series or 3d series	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn
2nd transition series or 4d series	39 Y	Zr	Nb	Mo	43 Tc	Ru	Rh	Pd	Ag	48 Cd
5d series	57 La	Hf	Ta	W	75 Re	Os	Ir	Pt	Au	80 Hg
6d series	89 Ac									112 Cn

+18

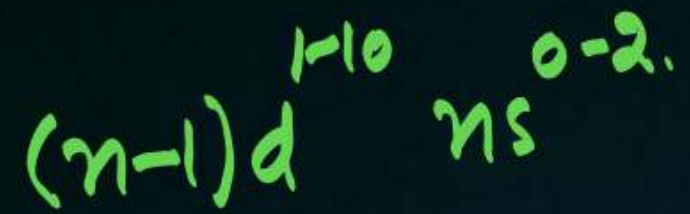
+32

+32

+18 +18 +32	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg
	Ac									Cn

Handwritten notes:

- ns^0 circled above Ni, with an arrow pointing to Pd.
- Arrows on the right side of the table indicating electron counts:
 - From Zn to Cd: +18
 - From Cd to Hg: +32
 - From Hg to Cn: +32



4th p. 3d series.



Ti (22).



Homework

Write the elements of d-block?

COMMUNITY Me
Port Karna.

GROUP	1	2											13	14	15	16	17	18
PERIOD	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	H Hydrogen 1.008																	He Helium 4.003
2	Li Lithium 6.94	Be Beryllium 9.012											B Boron 10.81	C Carbon 12.01	N Nitrogen 14.01	O Oxygen 16.00	F Fluorine 19.00	Ne Neon 20.18
3	Na Sodium 22.99	Mg Magnesium 24.31											Al Aluminum 26.98	Si Silicon 28.09	P Phosphorus 30.97	S Sulfur 32.06	Cl Chlorine 35.45	Ar Argon 39.95
4	K Potassium 39.10	Ca Calcium 40.08	Sc Scandium 44.96	Ti Titanium 47.88	V Vanadium 50.94	Cr Chromium 52.00	Mn Manganese 54.94	Fe Iron 55.85	Co Cobalt 58.93	Ni Nickel 58.69	Cu Copper 63.55	Zn Zinc 65.39	Ga Gallium 69.72	Ge Germanium 72.64	As Arsenic 74.92	Se Selenium 78.96	Br Bromine 79.90	Kr Krypton 83.79
5	Rb Rubidium 85.47	Sr Strontium 87.62	Y Yttrium 88.91	Zr Zirconium 91.22	Nb Niobium 92.91	Mo Molybdenum 95.96	Tc Technetium (98)	Ru Ruthenium 101.1	Rh Rhodium 102.9	Pd Palladium 106.4	Ag Silver 107.9	Cd Cadmium 112.4	In Indium 114.8	Sn Tin 118.7	Sb Antimony 121.8	Te Tellurium 127.6	I Iodine 126.9	Xe Xenon 131.3
6	Cs Cesium 132.9	Ba Barium 137.3	57-71 Lanthanides	Hf Hafnium 178.5	Ta Tantalum 180.9	W Tungsten 183.9	Re Rhenium 186.2	Os Osmium 190.2	Ir Iridium 192.2	Pt Platinum 195.1	Au Gold 197.0	Hg Mercury 200.5	Tl Thallium 204.38	Pb Lead 207.2	Bi Bismuth 209.0	Po Polonium (209)	At Astatine (210)	Rn Radon (222)
7	Fr Francium (223)	Ra Radium (226)	89-103 Actinides	Rf Rutherfordium (261)	Db Dubnium (268)	Sg Seaborgium (271)	Bh Bohrium (270)	Hs Hassium (277)	Mt Meitnerium (276)	Ds Darmstadtium (281)	Rg Roentgenium (280)	Cn Copernicium (285)	Nh Nihonium (284)	Fl Flerovium (289)	Mc Moscovium (288)	Lv Livermorium (293)	Ts Tennessine (294)	Og Oganesson (294)

78 — Atomic Number

Pt — Symbol

Platinum — Name

195.1 — Average Atomic Mass

57	58	59	60	61	62	63	64	65	66	67	68	69	70	71
La Lanthanum 138.9	Ce Cerium 140.1	Pr Praseodymium 140.9	Nd Neodymium 144.2	Pm Promethium (145)	Sm Samarium 150.4	Eu Europium 152.0	Gd Gadolinium 157.2	Tb Terbium 158.9	Dy Dysprosium 162.5	Ho Holmium 164.9	Er Erbium 167.3	Tm Thulium 168.9	Yb Ytterbium 173.0	Lu Lutetium 175.0
89	90	91	92	93	94	95	96	97	98	99	100	101	102	103
Ac Actinium (227)	Th Thorium 232.0	Pa Protactinium 231.0	U Uranium 238.0	Np Neptunium (237)	Pu Plutonium (244)	Am Americium (243)	Cm Curium (247)	Bk Berkelium (247)	Cf Californium (251)	Es Einsteinium (252)	Fm Fermium (257)	Md Mendelevium (258)	No Nobelium (259)	Lr Lawrencium (262)

Q. General electronic configuration of d-block elements.

a) $(n-1)d^{1-5} ns^{0-2}$

b) $(n-2)d^{1-10} ns^{0-2}$

c) $(n-1)d^{1-10} ns^{0-2}$

d) $(n-1)d^{1-10} ns^2$

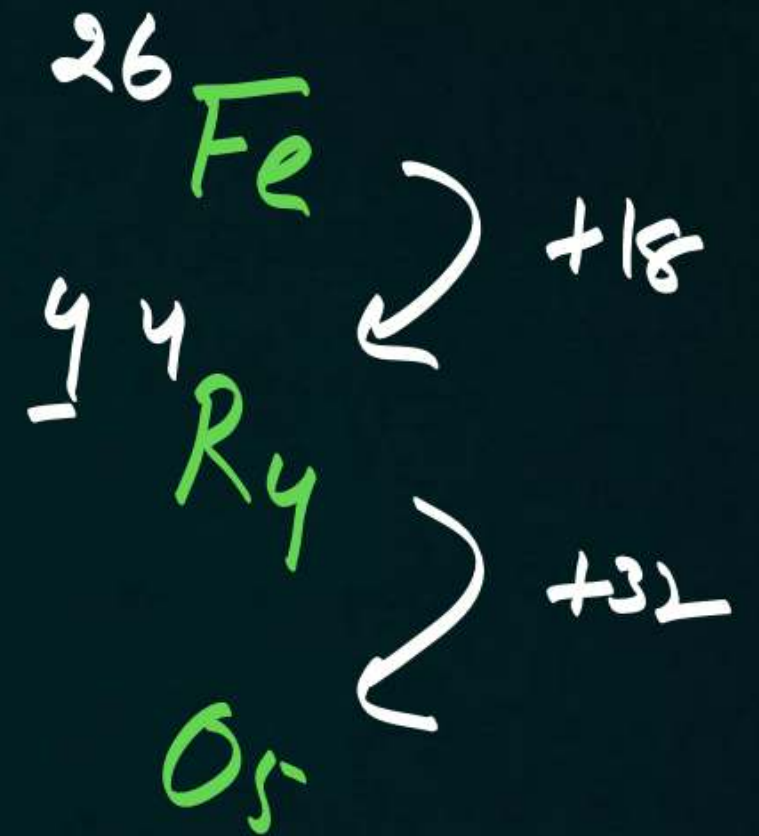
Q. What is atomic No. of Ruthenium (Ru)?

a) 34

b) 54

c) 64

d) 44



Q. D-blocks includes Group no.

a) 13 - 18

b) 1 - 2.

c) 3 - 12

d) None.

- ① Aafreena Fayaz
- ② UZAIR SUNAIL.
- ③ Tahangir

REVISION KAR 3 SONA.

Thank
You